Introductory Econometrics with Recitation — Quiz 1

March 10, 2026

Answer Key

1. (**10 points**) Let X be a random variable such that $E(X) = 6$ and $\mathrm{Var}(X) = 9$. Let $Y = -4X + 10$.

   (a) (**5 points**) Find $E(2X + Y)$.

   (b) (**5 points**) Find $\mathrm{Var}(3Y)$.

Suggested Answers for Problem 1:

(a) Since $Y = -4X + 10$,

$$2X + Y = 2X + (-4X + 10) = -2X + 10.$$

By linearity of expectation,

$$E(2X + Y) = E(-2X + 10) = -2E(X) + 10 = -2(6) + 10 = -2.$$

Grading Criterion

- **(5 pts)**: Correct answer.
- **(0 pts)**: Incorrect answer.

(b) First find $\text{Var}(Y)$. Since $Y = -4X + 10$,

$$\text{Var}(Y) = (-4)^2\text{Var}(X) = 16 \cdot 9 = 144.$$

Then

$$\text{Var}(3Y) = 3^2\text{Var}(Y) = 9 \cdot 144 = 1296.$$

Grading Criterion

- **(5 pts)**: Correct answer.
- **(0 pts)**: Incorrect answer.

2. (**20 points**) Data on monthly customer satisfaction index scores (measured on a 0-500 scale) for $n = 360$ retail branches in Brussels yield an average score of $\bar{Y} = 432.6$ and a standard deviation of $s_Y = 19.8$.

   Suppose the branches are divided into those with small sales teams (< 20 employees) and large sales teams (≥ 20 employees). The following results are found:

Team Size	Average Score ($\bar{Y}$)	Standard Deviation (s_Y)	n
Small	439.7	20.6	140
Large	428.1	17.9	220

   We want to know if there exists statistically significant evidence that the average customer satisfaction scores differ between branches with small and large sales teams. Assume all relevant conditions are met.

   (a) (**3 points**) Write the hypotheses in symbols and in words.

   (b) (**6 points**) Calculate the test statistic T and the associated degrees of freedom.

   (c) (**4 points**) Find the p-value and interpret it in the context.

   (d) (**2 points**) What is the conclusion of the hypothesis test at $\alpha = 0.05$?

   (e) (**5 points**) Construct a 90% confidence interval for the mean customer satisfaction score in the population of all branches.

Suggested Answers for Problem 2:

(a) Let μ_S be the true mean satisfaction score for branches with small sales teams and μ_L for branches with large sales teams.

$$H_0: \ \mu_S - \mu_L = 0 \qquad H_A: \ \mu_S - \mu_L \neq 0.$$

In words: under H_0 the two types of branches have the same mean satisfaction score; under H_A the means differ.

> Grading Criterion
>
> - **(3 pts)**: Correct hypotheses in symbols **(2 pts)** and words **(1 pts)**.
> - **(0 pts)**: Incorrect.

(b) Test statistic (two independent sample t):

$$T = \frac{\bar{Y}_S - \bar{Y}_L}{\sqrt{\frac{s_S^2}{n_S} + \frac{s_L^2}{n_L}}} = \frac{439.7 - 428.1}{\sqrt{\frac{20.6^2}{140} + \frac{17.9^2}{220}}}.$$

Compute the standard error:

$$SE = \sqrt{\frac{424.36}{140} + \frac{320.41}{220}} = \sqrt{3.031 + 1.456} = \sqrt{4.487} \approx 2.118.$$

So

$$T \approx \frac{11.6}{2.118} \approx 5.48.$$

A conservative choice for degrees of freedom is

$$df = \min\{140 - 1, 220 - 1\} = 139.$$

> Grading Criterion
>
> - **(6 pts)**: Correct $T \approx 5.48$ **(4 pts)** and $df = 139$ **(2 pts)**.
> - **(1-5 pts)**: Partially correct. Correct SE **(3 pts)**.
> - **(0 pts)**: Incorrect.

(c) p-value $= 2 \cdot \mathbb{P}(Z >= 5.48) \approx 0$.
Interpretation: Under the null hypothesis, the probability of observing a difference between the mean scores of 1.6 points or something more extreme is very close to 0.

Grading Criterion

- **(4 pts)**: Correct p-value range **(2 pts)** and interpretation **(2 pts)**.
- **(1-3 pts)**: Partially correct.
- **(0 pts)**: Incorrect.

(d) Since $p < 0.05$, reject H_0. There is statistically significant evidence that the mean customer satisfaction scores differ between branches with small and large sales teams.

Grading Criterion

- **(2 pts)**: Correct conclusion.
- **(1 pt)**: Partially correct. (e.g., correct rejection yet flawed reasoning.)
- **(0 pts)**: Incorrect.

(e) A 90% confidence interval for the overall mean satisfaction score is

$$\bar{Y} \pm t^{\star} \frac{s_Y}{\sqrt{n}}.$$

Here $df = n - 1 = 359$, so $t^{\star} \approx 1.65$. The standard error is

$$\frac{s_Y}{\sqrt{n}} = \frac{19.8}{\sqrt{360}} \approx 1.044.$$

Margin of error:

$$1.65(1.044) \approx 1.723.$$

Thus,

$$432.6 \pm 1.723 = (430.877,\ 434.323) \approx (430.9,\ 434.3).$$

Grading Criterion

- **(5 pts)**: Correct 90% CI setup and arithmetic.
- **(1-4 pts)**: Partially correct. Correct SE **(2 pts)**.
- **(0 pts)**: Incorrect method.